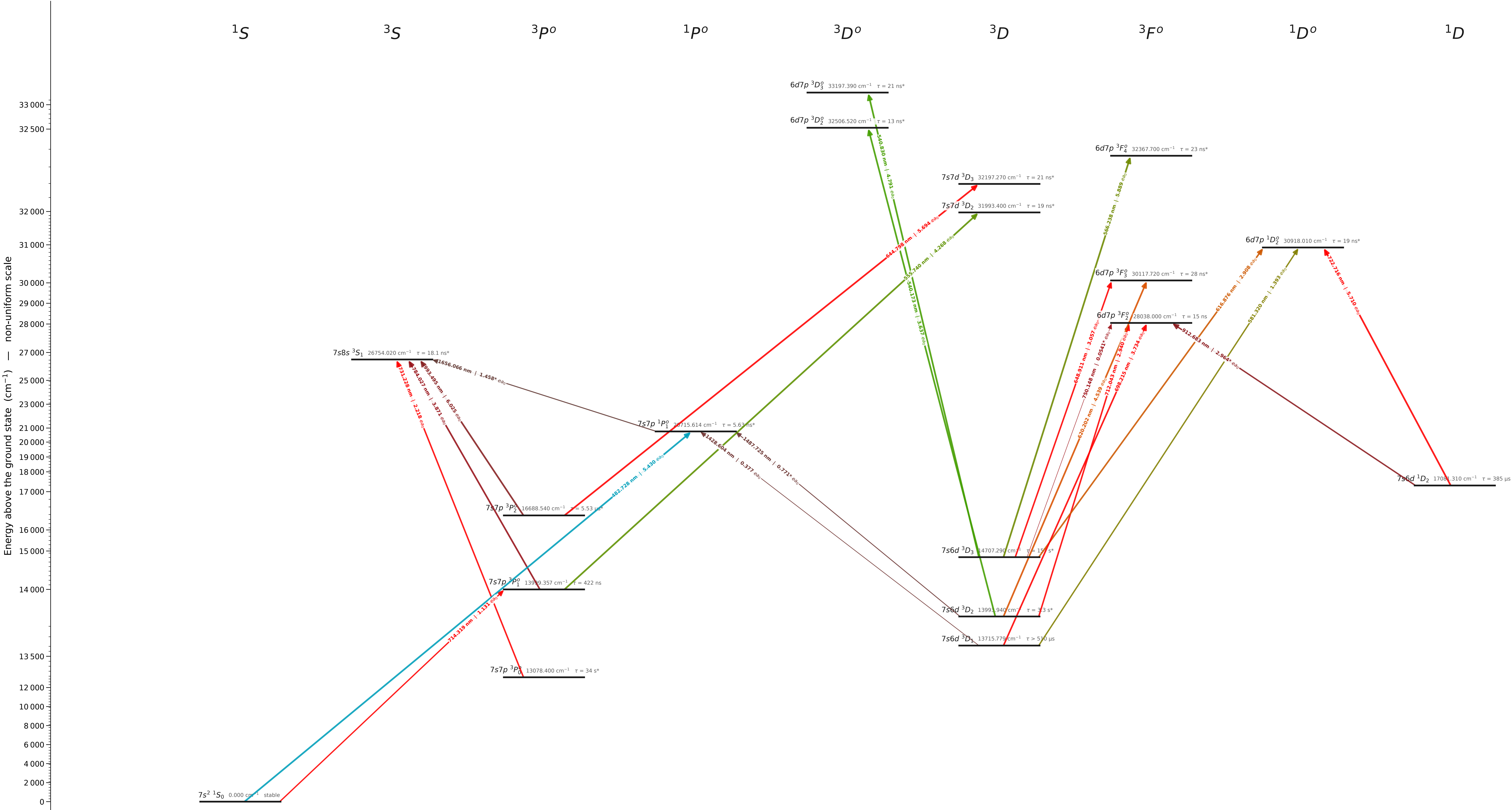
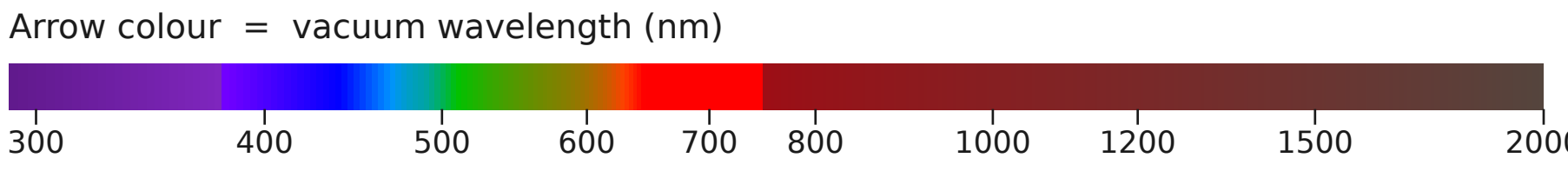


Radium

Ra (Ra I). Strongest classified transitions below 2 μm (NIST): 18 levels, 22 lines.

Ra⁺ ionisation limit 42573.36 cm⁻¹ = 5.27842 eV (234.889 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|⟨J||e\mathbf{r}||J'⟩|$ in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, *n* = 70.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	⟨J e $\mathbf{r}$ J'⟩ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
<i>7s</i> ² ¹ S ₀ – <i>7s</i> 7 <i>p</i> ¹ P ₁ ^o	482.7277	482.5928	621.038489(15)	5.430	1.770e+08	28.27	99.7	strong singlet line
<i>7s</i> 6 <i>d</i> ³ D ₃ – 6 <i>d</i> 7 <i>p</i> ³ F ₄ ^o	566.2383	566.0812	529.4458	5.889	4.300e+07	6.92	98.9	566.2 nm
<i>7s</i> 6 <i>d</i> ³ D ₁ – 6 <i>d</i> 7 <i>p</i> ¹ D ₂ ^o	581.3199	581.1588	515.7099	1.393	4.000e+06	8.377	7.6	581.3 nm
<i>7s</i> 6 <i>d</i> ³ D ₃ – 6 <i>d</i> 7 <i>p</i> ¹ D ₂ ^o	616.8757	616.7051	485.9852	2.908	1.460e+07	8.377	27.7	616.9 nm
<i>7s</i> 6 <i>d</i> ³ D ₂ – 6 <i>d</i> 7 <i>p</i> ³ F ₃ ^o	620.2020	620.0304	483.3788	4.539	2.500e+07	5.684	70	620.2 nm
<i>7s</i> 7 <i>p</i> ³ P ₂ ^o – <i>7s</i> 7 <i>d</i> ³ D ₃	644.7981	644.6200	464.9400	5.694	3.500e+07	7.579	73.5	644.8 nm line used for collinear laser spectroscopy at IS...
<i>7s</i> 6 <i>d</i> ³ D ₃ – 6 <i>d</i> 7 <i>p</i> ³ F ₃ ^o	648.9112	648.7319	461.9931	3.057	9.900e+06	5.684	27.7	648.9 nm
<i>7s</i> 6 <i>d</i> ³ D ₁ – 6 <i>d</i> 7 <i>p</i> ³ F ₂ ^o	698.2151	698.0226	429.369773(15)	3.734	1.660e+07	10.61	31	proposed spin-flip repump channel for the blue-slower sch...
<i>7s</i> 6 <i>d</i> ³ D ₂ – 6 <i>d</i> 7 <i>p</i> ³ F ₂ ^o	712.0434	711.8472	421.031151(15)	2.940	9.700e+06	10.61	64	proposed spin-flip repump channel for the blue-slower sch...
<i>7s</i> ² ¹ S ₀ – <i>7s</i> 7 <i>p</i> ³ P ₁ ^o	714.3185	714.1217	419.69016(3)	1.131	2.370e+06	0.3771	100	intercombination line
<i>7s</i> 6 <i>d</i> ¹ D ₂ – 6 <i>d</i> 7 <i>p</i> ¹ D ₂ ^o	722.7157	722.5165	414.8138	5.710	3.500e+07	8.377	66.5	722.7 nm
<i>7s</i> 7 <i>p</i> ³ P ₀ ^o – <i>7s</i> 8 <i>s</i> ³ S ₁	731.2283	731.0269	409.9848	2.218	8.500e+06	8.793	15.2	731.2 nm
<i>7s</i> 6 <i>d</i> ³ D ₃ – 6 <i>d</i> 7 <i>p</i> ³ F ₂ ^o	750.1476	749.9411	399.6446	0.054*	2.815e+03	10.61	0.4	750.1 nm
<i>7s</i> 7 <i>p</i> ³ P ₁ ^o – <i>7s</i> 8 <i>s</i> ³ S ₁	784.0270	783.8113	382.3752	3.871	2.100e+07	8.793	38.3	784.0 nm
<i>7s</i> 6 <i>d</i> ¹ D ₂ – 6 <i>d</i> 7 <i>p</i> ³ F ₂ ^o	912.6828	912.4323	328.473887(15)	2.964*	4.683e+06	10.61	5	proposed spin-flip repump channel for the blue-slower sch...
<i>7s</i> 7 <i>p</i> ³ P ₂ ^o – <i>7s</i> 8 <i>s</i> ³ S ₁	993.4946	993.2223	301.7555	6.025	2.500e+07	8.793	45.9	993.5 nm
<i>7s</i> 6 <i>d</i> ³ D ₁ – <i>7s</i> 7 <i>p</i> ¹ P ₁ ^o	1428.6041	1428.2136	209.8499(6)	0.377	3.300e+04	28.27	0.0231	repumper
<i>7s</i> 7 <i>p</i> ¹ P ₁ ^o – <i>7s</i> 8 <i>s</i> ³ S ₁	1656.0662	1655.6140	181.0268	1.458*	3.159e+05	8.793	0.572	1656 nm

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πc).

²²³Ra hyperfine structure (*I* = 3/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
<i>7s</i> 7 <i>p</i> ³ P ₁ ^o	1202.1(6)	-470.2(12)	F=1/2: -3593.00 F=3/2: -731.90 F=5/2: +1685.60
<i>7s</i> 7 <i>p</i> ³ P ₂ ^o	700(3)	688(4)	F=1/2: -2545.76 F=3/2: -2098.80 F=5/2: -780.11 F=7/2: +2270.93
<i>7s</i> 7 <i>p</i> ¹ P ₁ ^o	-344.5(9)	421.5(16)	F=1/2: +1388.12 F=3/2: -77.00 F=5/2: -411.38
<i>7s</i> 7 <i>d</i> ³ D ₃	536(5)	74(3)	F=3/2: -3169.26 F=5/2: -1885.68 F=7/2: -36.95 F=9/2: +2428.68

Measured A, B constants; sources in the data file.

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
<i>7s</i> ² ¹ S ₀ – <i>7s</i> 7 <i>p</i> ¹ P ₁ ^o	482.728	208-214	8485(8)
<i>7s</i> ² ¹ S ₀ – <i>7s</i> 7 <i>p</i> ¹ P ₁ ^o	482.728	209-214	8313(8)
<i>7s</i> ² ¹ S ₀ – <i>7s</i> 7 <i>p</i> ¹ P ₁ ^o	482.728	210-214	6003(6)
<i>7s</i> ² ¹ S ₀ – <i>7s</i> 7 <i>p</i> ¹ P ₁ ^o	482.728	211-214	5522(4)
<i>7s</i> ² ¹ S ₀ – <i>7s</i> 7 <i>p</i> ¹ P ₁ ^o	482.728	212-214	3256(3)
<i>7s</i> ² ¹ S ₀ – <i>7s</i> 7 <i>p</i> ¹ P ₁ ^o	482.728	213-214	2167(3)
<i>7s</i> ² ¹ S ₀ – <i>7s</i> 7 <i>p</i> ¹ P ₁ ^o	482.728	220-214	-21904(8)
<i>7s</i> ² ¹ S ₀ – <i>7s</i> 7 <i>p</i> ¹ P ₁ ^o	482.728	221-214	-25945(8)
<i>7s</i> ² ¹ S ₀ – <i>7s</i> 7 <i>p</i> ¹ P ₁ ^o	482.728	222-214	-28827(8)
<i>7s</i> ² ¹ S ₀ – <i>7s</i> 7 <i>p</i> ¹ P ₁ ^o	482.728	223-214	-32453(9)
<i>7s</i> ² ¹ S ₀ – <i>7s</i> 7 <i>p</i> ¹ P ₁ ^o	482.728	224-214	-35123(9)
<i>7s</i> ² ¹ S ₀ – <i>7s</i> 7 <i>p</i> ¹ P ₁ ^o	482.728	225-214	-38888(10)
<i>7s</i> ² ¹ S ₀ – <i>7s</i> 7 <i>p</i> ¹ P ₁ ^o	482.728	226-214	-41124(11)
<i>7s</i> ² ¹ S ₀ – <i>7s</i> 7 <i>p</i> ¹ P ₁ ^o	482.728	227-214	-43937(12)
<i>7s</i> ² ¹ S ₀ – <i>7s</i> 7 <i>p</i> ¹ P ₁ ^o	482.728	228-214	-46950(12)
<i>7s</i> ² ¹ S ₀ – <i>7s</i> 7 <i>p</i> ¹ P ₁ ^o	482.728	229-214	-50066(15)
<i>7s</i> ² ¹ S ₀ – <i>7s</i> 7 <i>p</i> ¹ P ₁ ^o	482.728	230-214	-53634(15)
<i>7s</i> ² ¹ S ₀ – <i>7s</i> 7 <i>p</i> ¹ P ₁ ^o	482.728	232-214	-59585(17)
<i>7s</i> ² ¹ S ₀ – <i>7s</i> 7 <i>p</i> ¹ P ₁ ^o	482.728	225-226	2236(15)
<i>7s</i> 7 <i>p</i> ³ P ₂ ^o – <i>7s</i> 7 <i>d</i> ³ D ₃	644.798	212-214	-887(4)
<i>7s</i> 7 <i>p</i> ³ P ₂ ^o – <i>7s</i> 7 <i>d</i> ³ D ₃	644.798	221-214	7001(7)
<i>7s</i> 7 <i>p</i> ³ P ₂ ^o – <i>7s</i> 7 <i>d</i> ³ D ₃	644.798	222-214	7781(4)

Shift = ν(first isotope) – ν(second isotope).